PP3632: Structure of A High-Speed Rail Corridor: Long-term planning implications for sustainable regional development along HSR Corridor

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Background & Introduction

- Spatial dynamics change with socio-economic, technological, and cultural factors, influenced by infrastructure development.
- High-Speed Rail (HSR) is a nascent global inter-city transport system, yielding positive socio-economic and policy effects.
- HSR enhances market access and business opportunities, fostering mutual growth beyond capital cities.
- Intensified economic networks and urban-regional development arise, forming Mega-City Regions (MCRs).
- Literature spotlights HSR's urban service sector impacts, while regional-scale effects and planning implications are emerging.
- Research gaps exist in assessing later HSR impacts and sectoral activities regionally, especially in emerging economies.
- The study's core focus is the role of planning in HSR infrastructure implementation, establishing a theoretical framework based on literature and the Mumbai-Ahmedabad HSR Project case study.

Aims & Objectives

- The objective of this research is to present the significant areas influenced by High-Speed Rail (HSR) infrastructure investments, contributing to the formation of a region of impact.
- This research aims to outline a theoretical framework for the configuration of HSR corridors, which has been developed through an extensive literature review and a comprehensive case study analysis of established HSR projects worldwide.

Material, Methods and, Framework

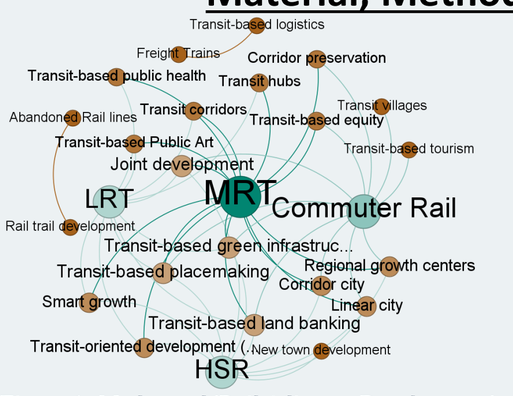


Figure 1: Modes and 'Rail Adjacent Development' concepts.

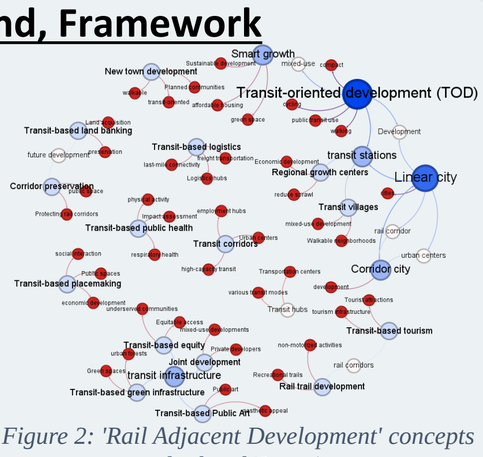


Figure 2: 'Rail Adjacent Development' concepts and related Domains

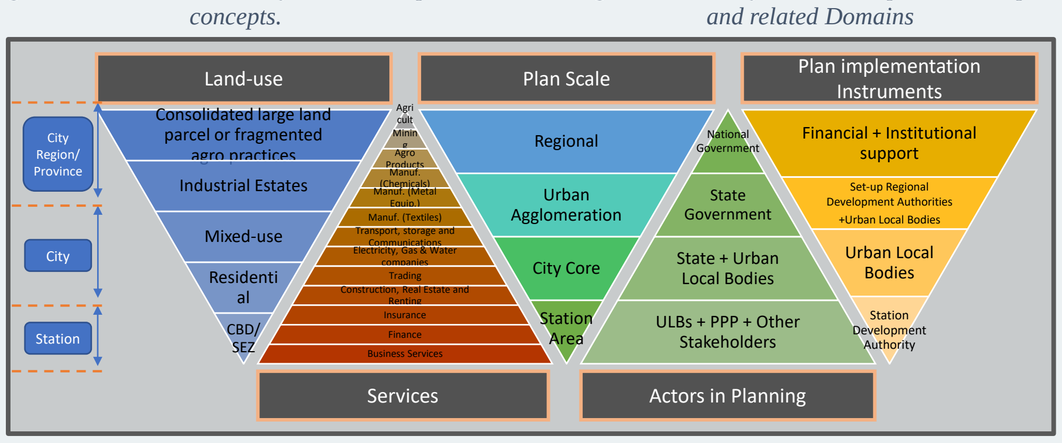
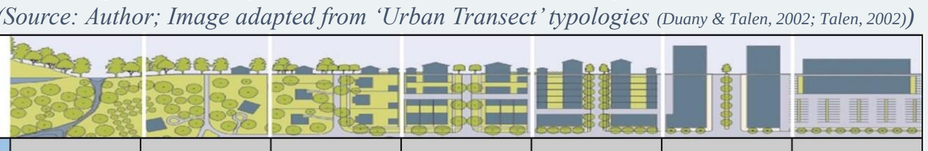


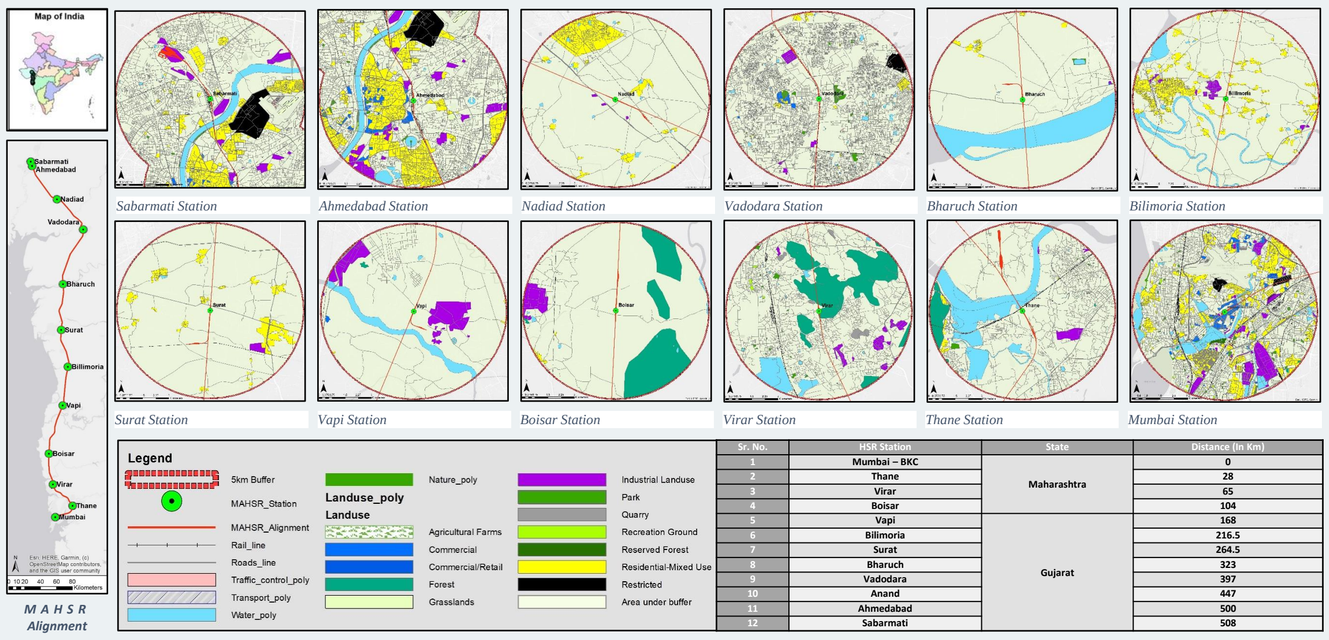
Figure 3: Illustration of a theoretical framework and components of a HSR Corridor (Source: Author)

Table 1: Parameters of urban development around HSR station on basis of 'Location' and 'HSR-Urban' interaction (Source: Author; Image adapted from 'Urban Transect' typologies (Duany & Talen, 2002; Talen, 2002))



Station Location	Natural	Rural	Sub-Urban	General Urban	Urban Centre	Urban Core	Special District
Type of Development for Station	New	New	New	New/Existing/ Integrated	New/Existing/ Integrated	Existing/ Integrated	New
Accessibility	Low/Nil	Low	Car/ Suburban-Commuter Rail/Feeder Service	Car/Feeder Service	Public/Car/Feeder Service/Metro	Public/Commuter Rail/LRT	Metro/LRT/Walkability
Type of possible Land Development	New-Township depending on the distance from closest CBD	New Town, Industrial, Tourism location	High with tendency to form urban Sub-Centre	High with tendency to form urban Sub-Centre	High Urban Renewal /Redevelopment projects	High Urban Renewal /Redevelopment projects	Low
Types of Investments it may attract	---------	Leisure & Tourism	Tourism, Industrial, Commercial, New residential townships	Commercial, New residential townships, Leisure	Activities associated to the closest CBD (Maintaining Industrial Ecology)	Knowledge-Innovation based activities	Localization of Firms, New Research Facilities
Stakeholders	Railway Operator + Landowner	Railway Operator + Landowner + Developer	Railway Operator + Landowner + Developer + Local body	Railway Operator/s + Landowner + Developer + Investors + Local body	Railway Operator/s + Landowner + Investors + Local body + Private organizations	Railway Operator/s + Landowner + Investors + Local body + Private organizations	Railway Operator/s + Investors + Local body + Private organizations
Proposed Stations in INDIA	Virar	Boisar, Vapi, Nadiad/Anand	Surat , Bharuch	Thane, Billimora,	Ahmedabad, Sabarmati	Vadodara	Bandra BKC

Mumbai - Ahmedabad HSR (MAHSR) Region



Sr. No.	HSR Station	State	Distance (in Km)
1	Mumbai – BKC	Maharashtra	0
2	Thane		28
3	Virar		65
4	Boisar		104
5	Vapi	Gujarat	168
6	Bilimoria		216.5
7	Surat		264.5
8	Bharuch		323
9	Vadodara		397
10	Anand		447
11	Ahmedabad		500
12	Sabarmati		508

Illustration: Mumbai Ahmedabad HSR Region (MAHSR) Overview: Links Gujarat, Maharashtra, and Union Territories. Key airports: Mumbai, Vadodara, Surat, Ahmedabad. National Highways connect towns to cities. State highways link tier two cities to lower-order areas. MAHSR Alignment: Runs along National Highway-8 (NH-8) and conventional railway line.

Structure of a HSR corridor

Analyzing the role of connected cities in safeguarding sustainable development policies is crucial for creating a resilient HSR corridor region, given HSR's global potential for growth.

- **Urban Networks and Corridors**: The concept of urban networks and corridors aligns with the rise of the network society and network economy, emphasizing open and horizontal economic and social relationships.

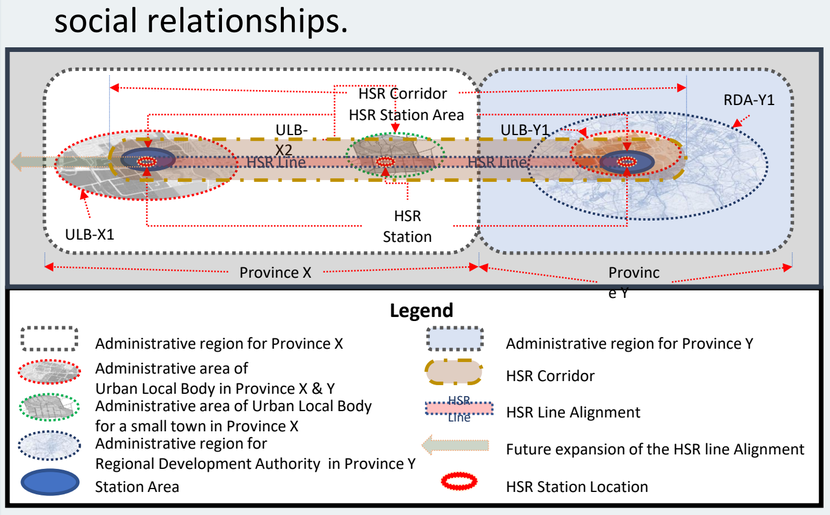


Illustration: Figure 4 demonstrates an interregional HSR Corridor connecting provinces X and Y, passing through cities X1, X2, and Y1, each governed by Urban Local Bodies (ULBs) or Regional Development Authorities (RDAs).

- **Mapping Spatial Organization**: Mapping a corridor's spatial organization involves considering not only traffic and infrastructure but also the functions of cities and towns within the corridor.
- **HSR Corridors**: High-Speed Rail (HSR) forms corridors that overlay administrative zones of urban activities, connecting multiple urban regions.
- **Importance of Identifying Corridors**: Identifying corridors is crucial for shaping spatial development patterns and administrative structures. It aids in safeguarding policies for sustainable development within the connected cities and towns.

Conclusion

The structure of an HSR corridor results in distinct city typologies, necessitates planning at regional, urban, and station levels, emphasizes synergy between transit elements for sustainable development, and underscores the importance of collaboration among cities along the corridor.

- **City Typologies in HSR Corridors:**
  - Cities in HSR corridors develop distinct typologies based on the concentration of business services, knowledge-intensive employment, manufacturing services, or agricultural practices.
  - This clustering results in sectorally specialized regions within the corridor.
- **Three Levels of HSR Corridor Planning:**
  - **Regional Level**: HSR impacts regional accessibility, changing mobility patterns and reshaping urban-regional systems.
  - **Urban Level**: HSR catalyzes spatial and urban transformation, primarily impacting the service sector and regional spatial development.
  - **Station Area Level**: HSR stations benefit both cities and operators, generating revenue diversification and internal city structure enhancement.
- **Importance of Synergy:**
  - Synergy between HSR and urban transit, paratransit, and station area development is vital for spatial and economic development.
  - Effective institutional arrangements are crucial to synchronize these elements and prevent fragmentation.
- **Collaboration over Competition:**
  - Cities in HSR corridors should collaborate rather than compete, with HSR operators facilitating cooperation among cities to enhance corridor development.

Acknowledgement & References

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